

Stakeholder Engagement

Risk Mitigation Strategy

How will your team reduce these risks?

- * **Rigorous Validation Across Diversity:** Test model performance separately across gender, age, region, socioeconomic status, language, and educational background. Measure sensitivity, specificity, and precision by subgroup. Set minimum performance thresholds for each group; do not deploy if performance is inequitable. Conduct qualitative interviews with users from each demographic to understand if model outputs resonate with their lived experience.
- * **Human-in-the-Loop Design:** Design the system as a screening aid, never a diagnostic tool. Model outputs are presented to trained mental health professionals or counselors for clinical judgment and follow-up. Users receive clear communication: "This is a screening tool to help you get support, not a diagnosis. A counselor will review this with you."
- * **Privacy by Design & Data Governance:** Implement end-to-end encryption for data in transit and at rest; anonymize training data; use differential privacy to protect individual records; limit data retention (delete after assessment); implement role-based access controls; audit all data access; obtain explicit, granular informed consent allowing users to opt out or request data deletion; partner with data protection officers and conduct regular privacy impact assessments.
- * **Culturally Informed Labeling & Validation:** Involve Nigerian and African mental health professionals, traditional healers, community leaders, and persons with lived mental health experience in defining what constitutes "distress" vs. culturally normative emotional expression. Ensure training data includes diverse communication styles (formal and informal speech, multiple languages, code-switching, cultural metaphors). Validate with end-users: "Does this model understand me?"
- * **Transparent Communication & Limitation Reporting:** Publish model performance metrics by demographic group; clearly communicate what the model can and cannot do; acknowledge uncertainty and error rates; explain how users' data is used and protected; provide plain-language information materials in multiple languages; train all users (patients, families, providers) on appropriate use and interpretation.
- * **Incident Response & Accountability:** Establish clear protocols for handling misclassifications (e.g., false positives causing unnecessary treatment; false negatives missing at-risk individuals), privacy

breaches, or reports of harm. Create a feedback mechanism allowing users to flag concerns.

Commit to transparency: publish incident reports and corrective actions.

- * **Equity-Focused Deployment:** Ensure the tool is accessible to low-literacy and offline users (SMS-based screening, USSD integration, paper-based alternatives). Test usability with intended end-users before launch. Partner with community health workers and local organizations to build trust and ensure referrals lead to actual care.

- * **Referral System Strengthening:** Before deploying the screening tool, map and strengthen mental health referral infrastructure. Partner with existing counseling services, community health centers, and NGOs to ensure flagged individuals can actually access care. Provide training and support to healthcare workers. Monitor referral completion and outcomes.

- * **Ongoing Monitoring & Fairness Audits:** After deployment, continuously monitor model performance, user feedback, and health outcomes by demographic group. Conduct annual fairness audits. Be prepared to retrain or retire the model if it begins to cause harm.

- * **Community Engagement & Co-Design:** Throughout development and deployment, maintain genuine partnership with the communities the tool is meant to serve. Involve them in problem definition, validation, and evaluation (not as subjects but as collaborators).

Success Metrics

What outcomes would define success?

Success is far beyond model accuracy. It encompasses:

Clinical Utility:

- * Healthcare workers report the tool increases detection of at-risk individuals compared to status quo (baseline measurement required).
- * Clinicians rate the screening tool as helpful, trustworthy, and aligned with their clinical judgment.
- * The tool reduces time to diagnosis for individuals who would otherwise remain undetected.

Equitable Performance:

- * Model sensitivity, specificity, and precision remain within acceptable clinical ranges (e.g., $\geq 85\%$) across gender, age, region, socioeconomic status, and language groups.
- * Performance is audited and reported publicly by demographic group.
- * No demographic group experiences systematically worse outcomes.

User Trust & Uptake:

- * Individuals voluntarily use the screening tool; qualitative feedback indicates they find it helpful, respectful, and culturally appropriate.
- * Families and community members support use of the tool.
- * Providers integrate the tool into routine practice (not abandoned after pilot).
- * Users feel their privacy is protected; trust in the system increases over time.

Referral & Treatment Outcomes:

- * Users flagged by the system who proceed to counseling show improved mental health trajectories (reduced symptom severity, improved functioning, increased help-seeking).
- * Follow-up with users shows they accessed care and found it beneficial.
- * Reduction in crisis presentations and suicide attempts among screened populations (long-term outcome).

Equity & No Harm:

- * Minimal false positives leading to unnecessary stigma or inappropriate treatment.
- * No false negatives among high-risk individuals (sensitivity prioritized over specificity).
- * Zero privacy breaches; robust incident response when concerns arise.
- * No evidence of discriminatory outcomes or unintended harm to marginalized groups.
- * Users report feeling respected and understood by the system.

Sustainability & Scale:

- * The system can be maintained and improved within local capacity (trained staff, infrastructure, budget) without ongoing external dependence.
- * Model updates incorporate new data and continue to perform equitably.
- * The tool is integrated into existing health systems; pathways for referral and care are sustainable.
- * Cost per screening is affordable; tool does not create financial barriers to care.

Public Health Impact:

- * Measurable reduction in undiagnosed mental illness in pilot communities.
- * Increased mental health literacy among screened populations.
- * Strengthened mental health referral infrastructure (secondary benefit).
- * Evidence that the tool addresses a genuine gap and is preferred over alternative screening approaches.